ActInf GuestStream #023.1 ~ Maxwell Ramstead, Dalton Sakthivadivel, et al.

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hello and welcome everyone it is june 13th 2022 and we're here in acting lab guest stream number 23.1 with maxwell and dalton we're talking about rebooting the free energy principle literature and their recent publication so i'll pass it to the two of you to provide just an initial introduction and some context for the paper before we jump into some specific parts maxwell first cool my name is maxwell ramstad i'm the director of research at the versus research lab where we specialize in probabilistic approaches to artificial intelligence um and of late most of my research has been on the physics foundations of the free energy principle and yeah very very much looking forward to having this conversation awesome and dalton sure um okay well i'm dalton uh i'm a mathematician and a physicist so right now i'm i'm uh at stony brook university in new york i'm also at the uh versus lab where we're doing some work on the free energy principle right now my research interests are about formal approaches to biological physics and mathematical foundations of complex systems and these kinds of questions one of the most interesting frameworks in this sphere is the free energy principle and that's uh how i've gotten involved in this literature awesome so maybe in the same order what brought us to this paper what was the gap that this paper stepped into and how did this team arise around the paper and i'll bring it up on the screen as well it's a good question well i i think the paper does a bunch of different things um the first i mean we've presented this as rebooting the free energy principle literature um i think uh yeah over the last it's been almost two decades now since uh the fep has been a thing uh i think the first paper was circa 2005 ish and the framework or the the field of study that has emerged around the free energy principle has changed a lot over the last two decades in no small part in terms of the self-understanding that we have as a community of practitioners working with these techniques and i mean as dalton pointed out a while back i think the last time that really the fep literature had been surveyed in a systematic fashion to kind of present where we were at uh where we where we are currently like the state of the art it was probably carl's uh monograph from 2019 january 2019 um i mean you know both of you were part of this international physics reading group that we had uh this was not this is not a simple uh piece to read and uh yeah since then um there hasn't really been uh basically a summary of where the literature is so on the one hand we were aiming to uh reboot the literature uh to to basically summarize uh the ways that the free energy principle had been applied uh in the literature and uh to also present a lot of the uh progress that had been made over the last three years a lot of which is due to uh dalton so i think that segues nicely into the dalton well too generous by far as usual but uh yeah so um maxwell raises a good point that there uh hasn't been really an effort to tie together all the different threads since probably 2019 one of the reasons why i mean that's that's a long time but it's not overly long however in the last couple of years there have been some pretty serious advances in the state of the mathematics and the physics behind the free energy principle so now you know three years later we're in 2022 it's not a long time but nevertheless there have been big developments that really beg this kind of a survey to be done and so that was one of the aims at least of the paper was to draw together all the different threads to unify them make them a bit more cohesive and and present the free energy principle as we're conceptualizing it today in a very top to bottom fashion so before we jump into the sections and the arguments and the claims in the paper which i'm sure will cover a lot of these updates i'd just love to hear what is the free energy principle what is active inference and how are they related i mean i i guess in the most general sense the free energy principle is a statement that says that we can understand the the dynamics of self-organizing systems as following um a path of least surprisal in the most general sense and then in that context uh it's good that you ask the question i think that there are at least two uses of the term active inference in the literature um the one that i've tended to use is a very broad sense which is just like active inference is specifying a path of least action um for what are called autonomous states so like the active and internal states of uh systems that can that are considered things under the free energy formulation active inference has also been used to refer to uh a class of specific partially observable markov decision process models um that have been deployed and you know most of the implement in silico implementations of the free energy principle rely on these i think this has led to a little bit of confusion in that you know when folks say active inference is a general theory uh it's able to account for self-organization that they're not talk they're not saying this specific class of pond vp models is effectively you know repurposable for everything um so yeah i i would say that this is the first sense that i outlined is the sense in which we say that active inference is a corollary of the free energy principle um and yeah i mean in its most general form the free energy principle is just a statement uh an alternative statement or restatement of the principle of stationary action where the action is defined as a surprisal in effect great dalton same questions to you yeah i think um maxwell pretty much um covered it uh so as i mentioned you know my background is actually not in the free energy principle not originally and it's not even in machine learning uh so i've come at this literature with a bit of a different um set of ideas and traditions and tools and so as a result i can't really talk about active inference and i'm not totally familiar with it but i can talk about the free energy principle um at length i'll try to be brief um so to me the the free energy principle is a statement that random dynamical systems uh do on average unsurprising things and and it's a very simple statement all it means is that if you have a random dynamical system and it's trying to get to some place in state space that is not surprising to it so some preferred region of state space maybe a particular phenotype maybe a particular morphological blueprint something like this the free energy principle is a statement of not only that that happens but also a set of tools to model how that happens how do systems get to these kinds of attractors and there's this very interesting way in which the the getting to that attractor can be read as inference and this is where these all these connections in literature appear to um things like active inference you can talk about how systems actually get themselves to that attractor as a perception an action problem and you can model it using this very famous uh variational free energy bound and you can show okay when the system is encoding some sort of belief about its environment and about itself that when it does this inference you have the minimization of some upper bound on surprisal so so the system is at least doing unsurprising things at the level of you know it's not exceeding some kind of intrinsic surprise that's some of the early formulations some of the more recent formulations take it even further and you say the surprising full stop of of some trajectory of the system is going to be minimized by the system so it's going to take on average the system is going to do what it expects to do which in this case is get to some region in the state space that it prefers to inhabit um what makes that not a tautology and what makes it something that we can actually operationalize and ask interesting questions about is the fact that there is this nice connection to inference there are all of these explicit mathematical tools that we can use to talk about surprising minimization not just as the kind of information theoretic statement of systems that aren't surprising or systems that aren't surprised due on surprising things it ends up a little bit more nuanced than that and so so to me the free energy principle is this whole host of results uh about random dynamical systems and especially random dynamical systems which are organized or which self-organize think that is uh how i would think of it and it's and it's got some physics and uh stuff that that we can actually talk about in that context awesome thanks both for those great answers i'm pulling up the paper and so many things in what you said i just wanted to follow up on but let's follow the main line of the paper and wherever you think there's some dots that could be connected or any questions arising or areas of future research or the connection to applications that you're working on or other people might think about getting involved in we can just take those little cul-de-sacs from the main line so i have the paper up on bayesian mechanics of physics of and by beliefs the two of you are the first authors there's also connor heinz magnus kudo barron millage lancelot decosta brandon klein and carl fristen any just overview notes on again like what led to this team or anything like that while we're on this first page um well from my perspective uh you know daldon and i have been talking a lot about this kind of stuff for the last year year and a half two years yeah i i won't i won't try to i'll try not to be too uh overly generous dalton but like i think yeah uh so some of dalton's work um has i think changed the way that we uh think about uh the fep uh in particular um what what dalton did in this great paper towards a geometry and analysis for bayesian mechanics is to show that the free energy principle and the principle of constrained axon entropy are actually dual to each other they are two sides of the same coin and i i you know i've been um i've been following this line of uh work for uh quite some time and i think it it foundationally changes the way that we think about the fep it moves it from like this um some might say exotic result from you know theoretical uh neurobiology to uh you know it connects it back up to um the foundations of contemporary physics effectively by showing that you know this thing that we we understand and almost take for granted uh you know maximum entropy inference procedures turn out to connect up to uh this free energy literature in a non-trivial way so i think the paper started out and it didn't end this way but i think we started off trying to write a reader-friendly companion to uh the work that dalton had done um but by the end of it realized that this was more um i think the dalton's paper towards geometry and analysis for uh bayesian mechanics pillars i think it stands uh on its own as a piece of mathematical reasoning so it is an exercise in a maximum of entropy inference math and it shows that you can recover uh some of the really core results of the free energy formulation just using the language of constraint maximum entropy in particular dalton approves the approximate bayesian inference lemma that had uh you know caused so much discussion in our for the last bit and so you know uh i think the the paper grew out of that um yeah the the team is something of a who's who of like the you know the the core contributors to this literature and just from a strategic point of view it was important for us to have like you know lance decosta carl fristen you know connor hines and folks that had been i think contributing to this core formulation and in particular to the development of uh you know this new framing of the free energy stuff a bayesian mechanics so yeah i think that the paper grew out of that and if dalton's paper is this kind of dalton's stent like the towards the geometry and analysis for bayesian mechanics is a kind of standalone piece of mathematical reasoning this is more about how this actually fits into the free energy principle literature so this is very much more a free energy principle literature paper and yeah we were hoping to do you know as reader friendly an introduction as we could uh there are actually two whole kind of introductory sections in the paper um but we ended up doing a lot of i think important work uh in that paper connecting up to the free energy principle i hope that helps answer me great answer and welcome lars would you like to give any overviews or notes before we jump into the sections of the paper hey thanks thanks for sending me the link there maxwell allow me on here um no nothing really to add there i'm just here as a spectator i have a few questions as we go down along and maybe i can throw into the conversation but um yeah thanks for having me on all right so the abstract begins with a really clear aim the aim of this paper is to introduce a field of study that has emerged over the last decade called bayesian mechanics and we're going to be unpacking that claim so suffice to say the abstract very clear and summarizes some of the core points that we're going to delve into now we're looking now at the contents what led to the sections the paper being this way and in this order what did you feel like it was important to include what did you feel like it was important to introduce and why well so if you look i'll i'll actually pull it up in front of me so that we uh we're all talking about the same thing give me one second and it's viewable in in large size on the live stream so yeah feel free to have your own copy up but i'll have whatever section we're looking at will be very visible on the live stream well uh maxwell is grabbing his copy um i'll just give a couple of contextual remarks so like maxwell described i mean originally i had released this set of results about what we call approximate bayesian inference the sort of mathematical basis for some of the key ideas in the fep literature up until that point anyway and originally it was meant to be a much shorter paper that just dissected some of those results and related them back to the fep in a way that was a little more concrete and a way that people who actually work on the ftp might might care a little more about um because of course the stuff that i wrote about is um a little far removed from the fep even though it's an object of inspiration for the results in the in that paper um there is a reading of that paper where it's formally pretty distinct from the fep so so originally that was the goal is just give it some extra physics intuition and some extra connection to the actual fep it became something a little bit different as we were building this out and we realized okay there are a whole instead of of new results um not only mine but also from uh carl's tnb group and from others folks and and they all really should be tied together nicely and part of that is the you know connecting the geometry and analysis paper to the actual fep part of it is actually talking about what we would think of the fep as today that shaped the table of contents a little bit because a lot of what we then wanted to go and say is okay we have theory of the free energy principle one of the key ideas in the way that we're viewing the free energy principle today is that it is just a bit of physics it is it is much like classical mechanics or statistical mechanics um it has a set of mathematical uh results that sort of hang around it that give it a nice foundation and it applies to specific systems and describes them in a specific way and and that was why we began with both an overview of what is mechanics what do we talk about in physics when we say mechanics and also section three is what is the free energy principle what what have we talked about so far in the literature and and what might we define as bayesian mechanics so what is what is the physics of beliefs or systems that carry beliefs so that that's sort of excuse me that's sort of um shaped the the first uh maybe 20 pages of the papers it's just building out the theory and the context for the theory afterwards is is a bit of a uh sort of change in um our uh line of attack but but you'll see it does remain very similar so so i won't uh get too far ahead of myself great any other notes maxwell or we can carry on i think that that's a that's a great summary i think uh yeah we we try to keep this as self-contained as possible as a paper so you know the the two you know introductory sections uh are meant to uh provide some kind of visual uh intuition for a lot of the more formal results that are discussed in the paper and in you know the the papers that it builds on um we have uh you know the artistic brilliance of brennan klein uh that's added uh to this paper so uh you know i think the the illustrations help um the idea was that you know based on dalton's work we we are formulating basically a geometry and analysis for the core results and the free energy principle literature so we thought let's leverage the geometric nature of a lot of these intuitions to really illustrate them and so that i think explains a lot of the structure and the paper ends on some philosophical considerations and hopefully clarifies some of the confusions or misunderstandings uh understandable misunderstandings uh that have arisen um yeah so i think that no that does really address uh i guess the last thing to point out is uh we we introduced this uh field of study uh in the last section that we uh we call g theory um which is i think the direction that we're going to be moving in for a while which uh is um yeah formulating uh the free energy principle in the bayesian mechanics in terms of maximum caliber so extending this duality between density dynamics formulation of the free energy principle in terms of that probability densities over states and maximum constrained maximum entropy to path-based uh formulations so path entropy is called caliber and you can dalton is one of the few world specialists in maximum caliber and yeah maximum caliber is a maximum path entropy so it's the same idea as maximum entropy but you're considering the entropy of paths we have had you know the free energy principle started off as a formulated in terms of paths after all it is a version of the you know the principle of stationary action it leads us to write down paths of stationary action and so the uh the ambition now is to connect these things up much more formally so that uh we can uh we can handle really genuine non-equilibria uh things like wandering uh markov blankets and uh yeah moving attractors and all that kind of stuff which find uh you know a home much more naturally in the language of maximum caliber did you did you want to add anything to that belton you're really the uh max cal expert um no i think that's uh that's a good account of um one of the key future directions is you know we spend the first half introducing bayesian mechanics we spend the second half talking about where bayesian mechanics fits into the rest of statistical mechanics and uh high energy physics by connecting it to max end and to gauge theory and this is where some of the geometric stuff comes in and then the the very last thing we do we sort of cap it off by saying okay now you know we've talked a lot about bayesian mechanics we've talked a lot about maximum entropy bayesian mechanics but there is in some sense much more natural formulation of bayesian mechanics in terms of dynamics because ultimately that's what we're thinking about is dynamical systems and the way that they evolve in time the way that they flow over their state space and there's there's a sense in which not only is it much more natural mathematically it just works easier it looks prettier it's so it's much nicer there's also a sense in which it's it's meaningless to talk about states of a process which is constantly in flux limiting ourselves to densities over states as exists in the last you know couple of years of the um which has in fact not always been the case some of the early stuff is written in terms of generalized coordinates of motion which are nothing but a particular way of talking about dynamics and therefore paths there's a sense in which the the recent um detour into states is very limiting not only mathematically but conceptually limiting and we can see that directly in the way that it is um handling things like non-markovian processes uh or processes that change with statistics that are out of equilibrium so so at the very end we say you know we've done all this work um the the best next step is to take all of this and and all this stuff that we've just done about maximum entropy basic mechanics and put it in dynamical terms and this necessitates passing to what we call maximum caliber which is uh also due to jains originally it's the principle of path entropy but uh eventually you know the goal is we would be able to write all of this stuff uh and in particular we'd have to be able to generalize what we have written into a language that's much more dynamical and and uh and max cal we already know uh has demonstrated uh an immense uh success in treating non-equilibrium processes so mathematically i mean there's reason to be very optimistic about what we will gain when we generalize the duality that we've talked about previously uh and so it really does seem like it's it's the ideal thing to do in the very near future let's jump into the sections so section one is the introduction and just one part that uh i wanted to highlight the footnotes are so informative there's some of these motifs like we set out to make it as simple as possible and four pages was our limit but then we ended up innovating this way so that's one funny motif and then the other funny motif that i see is um the super informative footnote often because maybe it's possible to just make the claim stand alone and indirect response to somewhere else so that's very exciting and interesting like footnote 1 covers a lot of the prerequisite fields like dynamical systems theory calculus probability and information theory so i found that to be very interesting because it helps understand where we can come from and what kinds of backgrounds we'd want to be learning about and then also a very important distinction of the two meanings of the word belief in the probabilistic sense of bayesian statistics or the propositional or folk understanding as you write from philosophy and cognitive science so um just thought those were two interesting footnotes and and reflected like part of the process and the discourse around the primary text and how that can be clarified in this format feel free to make any overall comments on that or that's just getting to page three well about the um the prerequisites it's a it's a difficult um it's a difficult balance to achieve in some sense right because um we don't want to be academic gatekeepers no one benefits from gatekeeping but these are also highly technical fields um so you know i've been joking for you know the last few months if you've never taken a derivative you're gonna have a bad time in this literature it's just not the kind of thing that lends itself to um surface engagement in some sense like it's very and especially with these um these new developments i mean uh there's dalton's work uh that basically shows that the free energy principle is dual to uh maximum entropy maximum caliber so there's this kind of deep connection to i mean effectively the the core of contemporary physics uh and there's the there are these new results uh that show that the free energy principle is asymptotically equivalent to the principle of unitarity from quantum mechanics uh you know there's there are these new quantum formulations of fep that have uh popped up yeah i mean navigating the space comfortably presupposes a lot and the good news is that it's hard for everyone so there's uh there's something to learn for everyone uh you know i spent the last uh three years of my life teaching myself like category theory and gage theory and stuff which turned out to be very useful uh you know connecting up with dalton make doing all of this work but now uh we're launching ourselves into quantum theory which i'm less familiar with so there's something to find difficult for everyone but um yeah it i i really do think that to fully appreciate going on here some technical prerequisites are important um otherwise we're led to making uh false assertions like i see this circulate a lot that dynamical systems approaches uh are contradictory to uh information theoretic approaches which is just nonsense from a formal point of view um yeah information metrics naturally re-emerge when you're considering for example uh how do you measure distances in state space effectively it's just yeah the the more uh mathematical background one comes with i think the easier it is to integrate into this um yes well indeed there's a lot of encouraging and good news and hopefully a community that also values participation and accessibility as well agreed important general points there um yeah before we move on i'll say something really quickly about this second book just because it's it's a remark that's actually pretty important to me in fact maxwell's the one who wrote it but uh i i didn't even realize but it is an important um clarification that when we talk about beliefs and an inference it's it's much more general than just uh an agent or a cognitive object the the manner in which we talk about inference and we talk about carrying beliefs is this this much more general sense of just two random dynamical systems that are coupled and and therefore reflect the statistics of each other so it could be a i mean generically this is a system estimating the statistics of an embedding environment the very important point is any system with any noise and in fact systems with no noise have statistics and so any system that estimates or couples to another system with statistics can be interpreted as mathematical inference that one system performs inference about the other the the sort of content that you place on that idea of inference is very case dependent so it's not the case that that stones for instance are cognitive but it is very much the case that stones reflect the statistics of their environment simply because these two systems are coupled and their statistics reflect each other and so this is what we mean in the sense of talking about agents that carry beliefs simply it is about objects that do this kind of statistical estimation which we think of as inference but it's it's certainly much more general than simply something like a reinforcement learning agent that actually stores representation of its environment um it is a representation but in a much more trivial sense in the general case i mean i would i would add to that by pointing out that um you know dalton was saying that we there were there were disadvantages to focusing on the density dynamics formulation which features markov blankets and the like up so marco blankets are great and we like them um but i think uh focusing on them so much in the literature over the last several years has obscured the fact that uh the the free energy principle is at core a statement about how you know as dalton was uh saying earlier random dynamical systems couple to each other so you know the markov blanket is not some i often hear and read in the literature um the markov blanket secludes the organism from its environment but it's not that's not that's not the idea at all uh the markov blanket is the set of states by which the organism and its environment are coupled and in fact the you know we made this case in a neat paper on a variational approach to niche construction several years ago you can basically swap the sensory and active states uh of your organism and you get sensory and active states of an environment it is fundamentally a story about how uh systems couple uh and synchronize with each other uh it's not a story about how things that exist or somehow secluded from an environment it's it's really a story about how yeah we uh we we become a model of our environment meaning that we are like in tune with engaged with and reflecting the structure of that environment this becomes very clear by the way when you look at the quantum formulations of the free energy principle if you take the asymptotic limit of the free energy principle so its behavior is time tends towards infinity um for classical systems that have space time individuation what you get is something like the good regulator theorem so i become as much as possible as i can like my environment uh you know modulo this space time separation i can't be my environment because i'm here and it's there in some sense and then what you get in that context is statistical mirroring but in in the quantum context when there is no space like separation between systems the fep tends uh towards um entanglement so systems that uh are related in an fep theoretic way at the quantum level uh become entangled that is precisely what the fep does conforming to the fep means that systems become maximally entangled so it really is not a statement about how organisms are separate from their environment it's a statement about how organisms are or things in general self-organizing systems individuate and couple simultaneously to their environments and how that's not a contradiction excellent um so as we rush through section one and head to section two i just wanted to pull out two really key quotes you wrote in this paper we discussed the relationship between dynamics mechanics and principles and then later on page five you write bayesian mechanics is specialized for particular systems that have a partition of states and go on to describe it more so as we head towards section 2 an overview of the idea of mechanics and indeed bayesian mechanics could you unpack a little bit by what you mean dynamics mechanics and principles sure um so mel andrews uh as everyone i think in the literature knows now wrote this great paper uh the math is not the territory um which challenged us to be very clear about what we meant really uh when we're talking about the free energy principle and active inference and in there i think mel makes a number of contributions but one of their most important contributions is to differentiate between the fep as an uninterpreted uh formal structure and the fep as it might figure in applications to model specific kinds of systems and uh mel points out that you know the a lot of the confusion in the philosophical literature has to do with the fact that the the primary ftp literature often doesn't differentiate the just the mathematical structure of the theory and then these models interpreted as representing certain features of a target system that we want to make sense of and that's important because falsification uh or verification empirical validation more generally um applies to one kind of thing but not the other right so uh the the free energy principle in so far it is as it is a mathematical theory i.e like a formal statement is no more falsifiable or subject to empirical verification than calculus would be right like you wouldn't you wouldn't try to falsify the axioms of calculus they do or do not apply informatively to certain kinds of systems so there's a level at which the core formal results of the fpp are have that status their mathematical results we can use this to uh you know do some interesting empirical stuff so uh what we wanted to do is to differentiate between three kinds of uh formal things in the first class of tools that mel was outlining namely we distinguish between dynamics mechanics and principles so the dynamics are descriptive they are basically a formal description of some systems behavior you might say that like galilean mechanics is almost entirely dynamics you know the you needed someone like newton to come around and say well there are uh general equations of motion that one can write down to account for sort of how these how this description plays out in some sense uh so we move from dynamics to mechanics uh in the same way that we move from description to explanation from a target phenomenon that is formalized in a certain way that we would like to have an explanation for to a set of equations of motion that really account for how those dynamics come to be and then principles if if mechanics tell you how principles tell you why so usually in physics we appeal to some symmetry or conservation principle in order to derive the mechanics or the mechanical theory for the the systems in question so what we're left with is principles mechanics and dynamics all of which are formal theories and once you have a mechanics you can apply it to study specific kinds of systems empirically and that's where empirical verification would come in so principles are not falsifiable they're just bits of math and you use them to write down mechanical theories which when interpreted in the right context and when the variables are mapped onto things that are of interest target systems then they become subject to empirical evaluation so hopefully that um you know uh clarifies a lot of the model around uh around the fep one of the reasons why we spent so long you know outlining the idea of mechanics and physics is that this isn't some exotic way of thinking about things this is really just physics just straight vanilla reasoning from physics um yeah i'll stop there dalton has probably some interesting things to add no i think i think you've covered it um yeah it's it's just that there are important formal differences in physics between what we would call a mechanical theory for some object and a dynamical description of some object and so in this tower of things you get ideas you get theories that sit at different layers in the tower this hierarchy ends up being useful in clarifying uh this host of different results in the fep and where one fits into the other so it was important to take the time to discuss it but that was that and that's the whole point of section two awesome so to also move quickly through section two on page nine you write we are concerned here with what we have called bayesian mechanics and then you launch into an example of mechanical theory operationalizing a mathematical and you provide some classical statistical formulations on page 10. so could you give us an overview of what the equations on page 10 are showing like what they do for classical physics and what is the analogy or what is the connection that this is preparing us to make with bayesian mechanics yes uh well in fact i have an entire paper uh that's about to come out about this just like building on the analogy um so i'll leave that as an easter egg for the audience watch out for a classical physics for the bayesian mechanic paper because i think instant classic it might be some interest yeah might be of some interest but yeah so on page 10 um we are just giving an example of the of this kind of formal difference so you have um some kind of principle in classical physics it is the principle that systems um are kind of lazy you know they there is some kind of uh an amount of energy that it takes to get from a to b and systems don't waste that energy uh so they take precisely the shortest path from a to b or the path of least energy in in the more technical sense of the transfer of potential energy into kinetic energy so if you have some amount of stored energy the transfer of that into energy of motion is minimized along the path that's a very um [Music] abstract statement and there's not a lot that you can actually do with that computationally all you can say is you know the systems systems uh try and get from a to b using the shortest amount of motion possible no extra motion so so if you see in figure one for instance uh there are these paths that fly off into sort of these these very strange fluctuations in the classical world those fluctuations don't exist because classical systems take this uh energy minimizing uh parabolic path in the darker blue what is computational about that is that principle of stationary action so the principle that you don't waste energy um results in a mechanical theory a law of motion for the mechanics of classical objects this is what we would call newton's law newton's second law in particular it's everyone knows f equals m a it's uh force equals mass times acceleration so the idea is that if you are such a system if you minimize the amount of energy that you then the acceleration of you must obey precisely the force being applied to you uh and um and so this is your your law of motion this is a mechanical theory the dynamics that come out of that are able to be discussed when you start putting in actual details when you plug in whatever force being applied uh sorry that's a dog you probably heard it uh what is what is the force being applied to the system what are the initial conditions for the system what is the mass for the system all this extra data starts giving you trajectories so an actual model of the system which is less explanatory and more descriptive and this is at the very bottom of the hierarchy where you get this dynamical theory that's got all this extra data in it so this is this is very quickly an account of the that hierarchy let's move into section three and figure so the caption is three faces of bayesian mechanics what are the three faces of bayesian mechanics and what is this figure showing it's often been suggested that there are like 17 000 versions of the free energy principle and that it's not very clear you know what the current version is and all that stuff and some of this is fair and some of it isn't i think some of it is fair because it is true that often in the technical papers not sufficient care has been given to you know ex making very explicit which assumptions we're making about the systems that we're interested in modeling um so what we tried to do in section three is to provide a comprehensive survey of all of the ways that the free energy principle has been articulated in the literature um and so it turns out that there are basically three main ways in which the fep has been applied which we're representing here so we've already alluded to the first distinction um the fep can either be formulated in terms of a probability density over states directly as a probability density over paths so over trajectories of states that we write in so-called generalized coordinates um and those are importantly different um so the the literature over the last i guess like circa 2009 2012 to 2019 focused almost exclusively on the uh density over states formulation and this is where you get um you know the approximate bayesian inference lemma that's played uh a pretty critical role in a lot of these investigations this is also where you get like states that play the role of markov blankets effectively whereas the original formulation of the fep was in terms of a probability density over paths so you're not you're not really so much interested in in the you know the the the dynamics of a system just in terms of how probable states are what you're actually interested in is you know entire trajectories uh that unfold over time and um the the most general you know paths formulation doesn't make a lot of the um assumptions that have been criticized over the last little bit so importantly uh we don't assume that the systems being described are stationary um we don't assume that uh you know and accordingly we don't assume that the system has a non-equilibrium a steady state and we're now returning to this formulation because it's less restrictive uh you know for reasons that uh the dolphin could probably detail uh better than i can um so then within the density over states um what you get is this kind of mode tracking or mode matching behavior where um i mean essentially what we're saying is that system that conforms to the free energy principle is such that it looks as if the average internal state is tracking the average external state or matching it and the the relevant difference in that density over states formulation is that external mode that the system is tracking can have dynamics to it or can also not um and it was important to distinguish these cases because a lot of the formal analysis so for example our colleagues at sussex released this great paper how particular are the physics of the free energy and they come to the conclusion that uh there's the fep doesn't really have anything interesting to say about the class of systems that they examined what is the class of systems that they examined they are looking at linear systems uh that undergo dissipation effectively so uh physically these are dampened springs right so uh to restate if you take away the mathematical uh baggage what they're saying is the fep doesn't have anything particularly interesting to say about uh the behavior of dampened springs so what does a dampened spring do well it starts off and then it dissipates back to a fixed point if you're burning right like just due to friction um so the mode towards which the system is flowing has no dynamics to it it is a fixed point it just dissipates back to a fixed point and what they're showing is that there's nothing interesting to say from an fep theoretic perspective about those systems but i mean the the counterpoint is sort of that there isn't interesting anything interesting theoretically to say about those systems there isn't any interesting uh external dynamics that uh the internal dynamics would be matching and a lot of these null results uh where we use fvp technology to say stuff about a system and there's nothing interesting to say look at systems that have fixed modes fixed external modes and so you know the the the idea is that you know the fep applies vacuously to those systems because there's nothing interesting to say um so to summarize my rant we uh we tried to summarize the main applications of the fep in the literature we identified um and strategically uh we point out where some commentators may have been led astray uh by focusing on one rather than the other so in particular a lot of this these ideas that the fep can't handle things like you know systems with moving attractors and uh you know systems that don't have like a single mode um that this these are understandable characterizations given that basically since 2000 from 2018 to 2 sorry 2012 to mostly focused on this middle class of systems so systems that uh we're describing in terms of the density over states uh that have a dynamic mode so it's understandable why people would focus their attention on that um but the fep in its most general form doesn't make you know perhaps problematic assumptions like stationarity um and it yeah it it also applies interestingly to systems that have interesting external state dynamics um so strategically i think that's what we were going for in this section um and i i believe this is the only place in the entire literature where this is reviewed so comprehensively in terms of this tripartite distinction um so you know i i don't mean to be disingenuous and to suggest that you the confusions that have arisen are only due to um you know just a lack of paying attention these things have not been expressed very clearly the assumptions have not always been stated very clearly and the the form of uh the path of least action in each of these things looks very different um so it we can also understand why it might seem like there are 17 versions of the free energy um but there aren't thank you maxwell as we speed through this section i'll just note that formalism 1 on page 15 is familiar and has to do with the flow of states in the particular partition and we've also seen the helmholtz decomposition shown in figure 3 in terms of a decomposition on into the vertical and the horizontal the gradient pursuing and the solenoidal isocontour aspect of flow those are reviewed in the citations that are provided here so let's get into the newer stuff in section four you're discussing some mathematic mathematical preliminaries on the maximum entropy principle gage theory and dualization but first yes lars moving on to section 4 section 3 was incredibly illuminating for me personally and i think the literature in general and having the separation out and there was a question that arose for me and working through this that i just wanted to throw out there um i don't want to distract the journey through the paper too much but in moving to this um you talk about on page um when the ftp is applied to density over states as opposed to path we assume a non-equilibrium steady state density and then on the next page page 20 you see we can view this this non-equilibrium study state density is providing a set of prior preference preferences which lead the particular system to look like it attempts to enact and bring about these preferences through action if you then move to the path-based formulation where there is no nest assumption how do you recover these preferences in that formulation or can you not use that formulation to describe systems that are doing actions that look like they have preferences or is there a piece that i'm missing does that make sense um i can take that one so um one of the one of the key uh deliverables of this third section is this three phases type thing right so the idea of the free energy principle and the mechanical theory that arises from it is we would want a mechanical theory a law of motion for bayesian inference or for systems that do basic inference and what it ends up being is approximate bayesian inference and and this comes in three flavors mode matching mode tracking or path track this is the the three faces thing so that i mean just like you have different flavors of classroom mechanics you have things that move continuously like satellites in orbit or you have things that move and then stop moving like a bull thrown through the air eventually returns to the ground and things that sit on the ground don't move unless they're acted upon so there are very different flavors of approximate basin infants and these three are in particular the kinds of things that we're thinking about so the reason why i give that context is because we can still imagine preferences on the paths of a system a system must evolve in such a way that it avoids certain areas of the state space given that it remains a system so uh you uh a human can't evolve in such a way that it dips too far away from its allostatic bounds there are certain set points and certain bounds around those set points in which i can oscillate but too far outside of those bounds and i'm in trouble likewise you know i can have something like a stone that kind of evolves through a state space the stone doesn't have preferences per se but it has a definition and and so we can read not um preferences but definitions about what a stone looks like in the same sense as we understand what a human looks like as certain preferences that guarantee cohesion so things that are definitional of a stone um guarantee that same kind of cohesion that just there's a type error if you would identify them but but you know mathematically they end up looking very similar and and what's interesting about that is now you can talk about preferences or you can talk about definitions along the path of evolution of a system a stone must stay roughly together it must stay in this sort of crystalline structure if you start uh hammering away at it you turn it into a powder this is something that we wouldn't think of as a stone so so in the same sense if a human doesn't eat for too long and and maybe starves to death it's a really grim example but unfortunately um it's one of the better ones you can imagine you know you're you're drifting too far away from this um this state of being satiated or having sufficient energy to maintain um cellular uh processes and the chemical reactions that sustain life and as a result you the the preference along your set of paths uh has been violated um so i think yes there is a sense in which even though we have forgone a nest density per se we still have a density over over paths not over states that this density still expresses something of a definition for the system or a set of preferences for the system without which we don't have a conceptualization of that system that's that's a very long answer to your question but but maybe it satisfies you no that's that's great thank you so much it was really useful that makes sense you you can kind of think about it as these constraints force systems to basically pick trajectories that are at the trial of uh you know a preference landscape or a constraint landscape if you want to think about it so you get the same kind of thing as dalton was saying but it's not the systems aren't assumed to be stationary so they're not um you know non-equilibrium steady-state distributions per se but there is something analogous uh and a lot of the uh future work in g theory is precisely about using the technology of maximum caliber to really make sense of this formally um let's move into section four so in section four as noted and as referenced to the previous publication there's a discussion that the fep is dueled to the constrained maximum entropy principle and then one of the ways that that's pursued in this section is through the lens or at least related to gauge theory we're looking at figure four what is gauge theory what does category theory duel and how is it being deployed in this situation um i guess i'll take that one as well uh figure four you said yes okay illustration of fiber bundles sections the big mega figures oh yeah okay yeah uh well mega fig i i recognize that name i i believe that's how we refer to it internally uh yeah well if you ask me um because i am a mathematician primarily if you were to ask me or if you were to ask a mathematician what is a gauge you would probably get a response roughly like well gage theory is the study of a particular geometric space called a fiber bundle so there is there is a type of space called the fiber bundle in geometry uh it looks exactly like that figure it's a bunch of fibers pasted onto um something on the bottom uh what we call a base manifold and uh and studying how do things behave a space like this how do they behave in the sort of fibers and how does that reflect some kind of behavior on the base manifold these are interesting geometric questions that arise in the study of fiber bundles and hence in what is called gage theory if you're a physicist i mean it turns out that fiber bundles are the natural space for talking about um classical uh gauge theories so the theory of certain high-energy particles um but it's but it's a little bit more general than that uh and so if you're a physicist this is what you think if you think about um gauge bosons uh but but this is only because fiber bundles are very natural way of talking about these things and so it is a little bit more general what we're talking about in the figure four is this idea that uh in this space of fibers uh you can kind of trace out um surfaces and that these surfaces are like probability densities over the base manifold and in fact they aren't just like probability densities they are so you can formulate probabilities over a state space as an object in a fiber bundle and therefore if you have all the technology available to you you can draw this very nice gauge theoretic interpretation of what it means to be a probability density and therefore because the object of basic mechanics is systems that do inference and so systems that estimate probability densities there is a nice gauge theoretic interpretation to bayesian mechanics that hinges pretty much on this picture what we would call the the gauge in this situation is the uh choice of this j function so you see you have several different surfaces uh three of them each of which comes with a different choice of j uh so the the idea of choosing a function in the fibers and therefore choosing a shape for your surface this is what we would think of as a gauge theory and and that's why i went down this route in the geometry and analysis paper because it ends up drawing a very nice picture of what it might mean to engage in constrained inference awesome and could i give a pedestrian or so the the bystander yeah the the more the less technical version of it is uh so gage theories and dalton correct me whenever i say something that's wildly or even not so wildly inaccurate but gage theories are basically how one would how one articulates symmetries or conservation laws and contemporary physics so most physics relies on the idea that some transformations that are of interest like the ones uh described by maxwell's equations uh have some kind of symmetry or or that concern i think a slightly better way of going about it would be to say that you know modern physics is broken up roughly into stuff and forces yeah and forces act on stuff and stuff gets acted on by forces so the interesting thing in gage theory is you can talk about how a definition of some stuff the shape of some stuff is intimately dependent on the forces acting on the stuff and so this is where we get this idea of we have a a particular shape to our surface we have some you know probabilities over our states but that this is actually defined by the the the choice of gauge this this j function so our assignment of probability two states is actually determined not by probability but it's determined by the constraints on those states certain states are preferable certain states are not unpreferable states get low probability preferable states get high probability and our notion of preferability is encoded in the choice of j it's a function that constrains different states so constraints in this case are like penalties uh so our assignment of penalties so our choice of force is actually what shapes our assignment of probability that's ultimately um i think uh well that's a slightly different way of viewing um the geometry of the situation is gage theory is all about these kinds of what we would call covariance relationships covariance meaning literally two things varying together if i choose a force i'm i'm forced to choose some stuff uh and likewise if i choose some stuff that necessitates uh an underlying choice of force so gauge covariance ends up being very helpful in situations like this thanks let's see if we can touch on some of these last figures in figure 5 we see a vector field on a curve that's reminding us of that sort of baseball diamond shape from earlier with the path of least action and in figure 6 a probability density is generated by level sets of the constraint function j so either figure five or figure six whichever you feel fits in best here what is being shown and where were you going with it how is it connected to alternative phrasings yeah i guess i'll just cut to figure six which is kind of about this covariance thing so if i choose particular functional form for j in this case it's a penalty that is proportional to distance i would expect my assignment of probability to fall off roughly like a distance function so it should it should be probability of a state should decrease as the penalty on a state increases so in this case the probability of a state should decrease as i go away from some central point and it should do so symmetrically and and so we would expect a gaussian density from and it is indeed the case that uh you can understand the covariance as a kind of projecting down and then lifting back up and what do you get when you do that well the stuff in the equation the probability uh ends up being defined by where you take all of the sort of rings on the the j function you project them down you get circles on the base manifold when you lift them back up they're rings and so we know a symmetric probability density with a whole bunch of rings is just a gaussian probability density so so this this figure is kind of capturing um what does this covariance relationship actually look like how do we actually build stuff out of forces and and it ends up being the case that it's this very simple recipe of you push everything down and then you lift it back up and the lift is e to the j so the so you place j on to x uh that's the first step your second step is then you hit it with this exponential function and e to the minus x squared is of course a gaussian density so that's all that figure 6 is depicting it's just zooming in on one of the objects in figure 4. great and i thought um on page 34 also the striking result that the splitting of the flow of a system into vertical and horizontal components under the constrained maximum entropy principle is isomorphic to the helmholtz decomposition of the flow of the autonomous partition of a particular system so for those who are caught up with the way that the helmholtz decomposition was used for example in the papers that we discussed in live stream 32 with markov blankets and chaos stochastic chaos this is augmenting it or transposing it into another formalization um and moving on again quickly um it's worth pointing out that there is uh there are some significant advantages to reformulating things in in terms of maximum entropy um uh i'll i'll let you know delta and speak to the more technical uh parts of this but i guess one um one thing that pops out to me more politically is that uh maximum entropy and constrained maximum entropy i mean they are just core parts of the physicist toolbox the fact that we are able to reproduce the helmholtz decomposition in terms of the way that flow splits up in the gauge theoretic formulation i think is quite significant it gives us an entirely independent line of reasoning arising at the same results so uh you know especially due to some some work by martin beale and colleagues a few years ago we were not we were no longer sure whether some of these results uh obtained you know the there had been doubt thrown on to the derivations of the approximate bayesian inference lemma in some papers uh life as we know it from 2013 i believe is the paper that they targeted uh one way of reading this is that independent of all these analyses and independent of whether bl and colleagues are correct about their criticism of the 2013 paper this is an independent derivation that just follows from uncontroversial mathematics and we're able to re-uh re-derive basically the the main core uh portions of the formal results uh that carl had derived so the approximate bayesian inference lemma the helmholtz decomposition um and i mean in some sense the the magic is that we understand now why these arise to begin with uh there's uh i think they're the um the maximum entropy interpretation um in some sense especially in the the gauge theoretic formulation in some sense tells us well why why does approximate bayesian entrance work just full stop and these sets of results explain you know effectively why this has to be the case only my facilitator hat prevents me from being that excited section 5.4 also explored more in the previous citation so the four things that i wanted to just touch on as briefly as people wanted as we close are the three philosophical areas that are introduced in section six and then of course just the appetizer of g theory so in page 38 section 6 the philosophy of bayesian 6.1 addresses clarifying the epistemic status of the fep so what were you aiming to bring people's attention to and address in 6.1 with this philosophical question well i think this basically summarizes the point about dynamics mechanics and principles that we were making earlier um you know so the fep has been described variously in various and sometimes very different ways even by like its core proponents so carl uh introduced the fdp to the world as a theory of cortical function circa and then as a unified brain theory um in 2010 which led a lot of people and i think not to this is in bad faith it led a lot of people to say well okay so this is a and so it should be making predictions it should be empirically verifiable uh so i think you know the the way that we talked about it led to a lot of uh confusion and it's been described as you know a framework whatever that means by myself included right so meia koopa we called it the active inference framework for a long time without really being specific as to what that meant we kind of meant all this this cluster of results that kind of agglomerate around the fep raja and colleagues have called this a trick uh kind of like a variational auto encoder trick in machine learning so it's a it's a trick if you can write a markov blanket for a system that you can make you can write down a model that looks as that says that the system looks as if it's performing bayesian inference it's also been called a physics a physics of sentient systems a formal ontology um so the point is to say well everyone is kind of right you know these are all in some sense valid assessments and mel is really the one who clarified things initially i think by distinguishing formal structures from interpreted empirically evaluable uh models um yeah i think the first part here is to say well we can make sense of this model by appealing to this tripartite distinction that we've outlined so by distinguishing dynamics or descriptions formal descriptions of some behavior that we want to get to mechanical theories that explain how and principles that explain why and then what we're calling i think empirical applications uh which are basically uh these formal models plus uh an assignment or an interpretation or evaluation of these models in terms of the features of target systems that we want to explain yeah so when they're applied in that way mechanical theories become empirical theories uh as we write in the ordinary sense but yeah so hopefully this kind of clarifies everything around falsification how this thing is used and importantly why this is just standard physics i mean this isn't this isn't all that spooky in terms of the use of mathematics to make sense of physically relevant aspects of our universe so that's what we were trying to do in section great really clear 6.1 it reads to me like it's the tip of an iceberg with a nice landing and take off strip and yes it's a big iceberg but it is an iceberg and there is a tip that makes sense 6.2 on page 39 elon vital what does this french linguistics have to do with the fep maxwell well uh dalton might have some pretty interesting remarks to make but just briefly i mean um i find it kind of ironic that uh you know the people who are most critical of fep right now like folks from the inactive tradition and folks from the ecological tradition sometimes frame their own research is trying to you know struggle against dualism in some sense but our belief is that they are intrinsically dualistic frameworks um so elon vitale is a french term with vitalism i don't know how familiar you are with like the history of biology but before the 20th century in the uh discovery of dna most serious uh physicists believed in something like aristotle's entire key so like that you you had to have some kind of physical mechanism for propositiveness and so ella guitar was this kind of vital force that uh they believe the vitalists believe animated living systems and differentiated them from non-living systems like stones and rocks we believe that one of the consequences of the perspective that we're adopting here and building on you know especially the work of mike levin who's done amazing uh made amazing contributions in this field is that it self-organization and cognition and life and these these things that we like to point out as being distinct from inert matter are actually are more on a continuum uh the you know the bayesian mechanic says that um basically anything that's a thing will exhibit some features of what we think of as cognition um so uh it is a massive massively deflationary and absolutely resolutely monistic perspective that we're advancing here yeah it was just you know trying to look for the boundary between living and non-living systems is bound to fail in some sense there is a continuum these processes that we're interested in are also evidenced by rocks and by tornadoes and by crystals and it is a question of degree uh to which systems uh act as cognitive systems i'm sure dalton has more interesting things to say about that yes on 6.2 what else would you add there dalton well maybe not more interesting perhaps more interesting things to say well yeah i mean i think this section is is mostly just um a recapitulation of footnote two in a meaning you know we we made this statement that inference per se representation of some statistics of something else is this very broad very general thing and any system that is not a closed system does inference so what that means is you know under the free energy and at least what we've written in that paper was you know what we've worked out so far uh there isn't uh a way to talk about cognition versus non-cognition it's more of a spectrum of cognition there are certain systems that are more competent this is something that carl might call a temporal depth so there are systems that are very good at planning and executing actions and there are systems that are not you know so there's a very clear difference between humans and stones for instance but that difference isn't in one does inference and another doesn't it's that one does a bit more complicated inference it has uh better or more accurate representations or more representations full stop maybe uh and it's better at using those representations to to do action but both stones and humans are estimators of the statistics of the processes around them simply by mathematical construction and and so there's this very uncontroversial statement or at least it shouldn't be a controversy that systems in the very most general sense exhibit this kind of representation-based inference that we would typically identify as cognition so that was the motivation for talking about that and then maxwell tied a very nice bow on it with this uh this great history of a debate around very similar questions uh but yeah it was this idea of a spectrum of cognition awesome and citation 92 there is the 2018 paper answering schrodinger's many years ago the other maxwell at all who raised that question and sparked really a phase transition in the discourse and that was also like one of those it was one of the papers that really drew me in and got me excited and so it's amazing to see what other questions and answers have been risen 6.3 and then g so 6.3 is a slightly longer section and it's on maps and territories so yeah what is that i'm really glad i'm really glad that we were able to write this so i i never found this particularly confusing but i i i i it has led to a lot of confusion um so in particular there's this entire paper by thomas van ness that just is just confused about this issue uh i forget what it's called as the living life modeled or what anyway um folks have said well so is is the free just just a description that we are uh you know a model that we are using to describe interesting features of the world or is it saying that um you know self-organizing systems actually are models of the world so you know i never particularly found this problematic because i so the this phrase uh in in the paper i think sums it up correctly uh the free energy principle is in some sense a map of that part of the territory that looks as if it is a map um and so uh thomas metzinger actually retweeted our paper and pointed uh called this nested representationalism which i found yeah there are these kind of two distinct levels at play but there's no foul play here because fep model is very explicit about which parts of the formalism pertain to our map of the territory and which parts of the formalism pertain to the map that self-organizing systems are constructing according to our map um yeah so i i i think it's important to point out that you know there isn't a contradiction here um that there are just these two probability densities that are at play in the construction the q density and the p density and you can understand the the generative model or the p density as basically uh our model of uh self-organizing system in terms of uh yeah joint probability density uh over uh it's it's state space or potentially it's a space of paths in that sense you know um the self-organizing system being its own best model uh really means we have a model uh we can explain the system's dynamics and the the system's uh potential uh is is formalized uh you know as a as a generative model as a joint probability density it just so happens that part of the variables that are described by that joint probability density uh play the role of parametrizing additional probability densities over another subset of variables described by the generative model so there are really these two i think unambiguously different senses of being a model that are at play i don't think this is a sleight of hand and hopefully we have clarified yeah what what exactly is at stake here um i think there are some nice connections to the maximum entropy uh formulation so as articulated in the paper at least uh and no he you said you said as much in discussion today um i think you by by flipping to the perspective of uh maximization of entropy we we effectively recover our own position as modelers trying to model the system the the the prism through maximum entropy is uh adds this additional clarification that uh yeah we can entirely recover the perspective of a model or using maximum entropy modeling to model a free energy type system by invoking this duality of perspectives and i think that's really important then there's this other piece here which says that the the maps that we construct in fep and in max end uh both adhere to a kind of epistemic minimalism so maximum entropy inference is basically uh saying i'm going to uh well given some data set uh the uh probability density that most that is uh the one that best explains the data set is the one with the highest entropy what this means is that we are uh building in to that uh probability distribution as little prior information as possible it's like starting from the point of view of maximal ignorance given a few constraints what is the underlying probability density the fep has a similar you know starting point for maximal ignorance which says i can't go beyond the blanket like i i am stuck with the data that i'm generating and uh with the way that i probe the world this data generating process to generate my data and th there is no going beyond that you can't go beyond the blanket unless you break the blanket but then you're just introducing a new blanket so for example um you know i i can't see in your brains um i could put you in an fmri machine and break the blanket but then i'm i'm really introducing another another blanket you know which is this new set of data that i have to continue making inferences from um i mean the then the the fep and accent kind of emerge as uh these epistemically um minimalistic or um they kind of emerge as the way that you would construct a map of the part of the territory that acts as a map from a perspective from from a starting point of maximal ignorance great and i think that's connected in the active inference textbook to figure 9.1 the meta bayesian perspective and there it's not described in the context of maximum entropy but it's something very similar with this nested representationalism on any other philosophical notes before we go to section seven yeah that's exactly right that's exactly right yeah yeah i love this book um yeah is that anything to add before we launch into g theory no i think you raised an interesting point um so we can talk about when we use the fep to model a system what we're doing is forming a model of the system where the system forms a model of its environment and so there is this nested representationalism what's interesting about the maximum entropy point of view is you can um kind uh dualize it you can you can take the sort of mirror image of this thing and you can ask okay well we're not interested in a model of the system modeling something all we want is a model of the system so what does the system look like when it's creating that model not what does the system's model look like so this is this is the dualization aspect of things and so you can talk about and when you dualize you can talk about things like constraints on agents rather than the predictions and actions of the agent so it's a bit of a different viewpoint it's just emphasizing different levels of the nest so to speak which is which is kind of interesting because you do get both of these perspectives for free as soon as you commit to this relational symmetry of an observer in an environment and an agent in an environment and how those two things do model each other um and that's what you get when you dualize things uh but that's yeah that's that's it for me on that section i think maxwell pretty much covered uh a lot of the important points i i mean i i will say that you know this this probably is not obvious until you really look at the mathematical formulation uh and notice that there are two you know as i as i wrote in that paper from a few years ago the fep is a tale of two densities um and you really need to uh like take take a look at the math to understand that there isn't a this isn't like a case of reification this is just formulated so it it it isn't so simple that you could just put it in a direct and a simple graph and say hey look here are these two but definitely if you take the time to work through the um the formalism there are clearly these two different senses of uh being a model that are at play um that are cleanly distinguished and i think are actually not all that awesome it's almost like thinking through other minds highlighted some of the social consequence and theory of mind and behavioral aspects of certain types of cognitive entities but systems thinking of through about other systems or infers thinking through other infers infra ants that type of dual view from the inside view from the outside is again what you're suggesting is enabled by that kind of a commitment to these foundations of course it could be a whole other guest stream and perhaps should be on g but as we close here what can we say today on 613 2022 about g theory and what is exciting you all about how you're going to continue forward do you want to take this one dalton uh yeah sure uh so i mean like i mentioned earlier on um there's already a lot of really promising stuff in the literature about patented girls and path entropy and the kind of power it has to deal systems that we would uh normally kind of balk at and and so it's it's already very promising there's reason to be optimistic about how this extension will benefit us in the future so g theory itself is is um maxwell coined this term to describe the uh extension of this duality from mode matching and accent max cal which will involve um precisely this we'll try to formulate um the free energy principle in its full generality as dual to uh entropy in its full generality on paths and so forth um and and so the reason why this is so exciting is because already there's a very clear path towards talking about um systems that are non-stationary so systems whose statistics change more complicated forms of inference uh more complicated forms of constraints and preferences and so it seems like already it's a much more faithful description of the kinds of systems that we want to talk about and extends to things like um more complex systems uh things like humans potentially um so so this is really the uh i guess the the end goal is to do this dualization to now build an fep dual model of something like a brain which we know models its environment in this very complicated way and so the kinds of representations it does are very complicated so how do we model that that that should uh arise out of maximum caliber and and therefore out of g and in principle you know eventually we'll have a very faithful uh very foundational picture of complex systems but this is a very um long-term goal in the short term we just want to get the maths to work to connect that a bit to lars's uh question earlier uh what does it mean to have a non-stationary density over paths well i mean what we're interested in is the way that the density changes as a function of the dynamics so that you really have like these moving surfaces over a space of paths or space of states and to really capture all of that kind of time dependency yeah requires moving from uh accent to maxcal uh from states to paths and um all of these kind of more fine-grained wandering blankets and moving attractors and all of that good stuff should fall fairly naturally out of yeah g theoretic formulation of this i like how within fep as we know it we have variational free energy f and then expected free energy takes us into the future and planning and that's g and then we see a little bit of like free energy principal f and then g is the next letter so that's a funny little would know duel analogy um if anybody has any closing thoughts it would be a great time for that but just wanted to say that there's a lot of appreciation in the chat and a lot of excitement around this work so we hope that you just continue acting and inferring and serving that's that's kind of you to say uh yeah thank you very much for the invitation to discuss our work this has been great uh and yeah we uh we look forward to uh you know exploring the future uh with this whole community and um thanks for your time your attention your energy very very positive always a pleasure to be on actin flab and uh yeah thanks for the guest stream yes i can only echo that uh it's been a pleasure so thank you thank you thank you maxwell and dalton at all thank you lars for joining and until the next time bye may the gradient descend ever in your favor my friend and with yours slash ours take care